REWARD ESTIMATION OF RISK SENSITIVE AGENTS THROUGH GRADIENT BASED INVERSE REINFORCEMENT LEARNING

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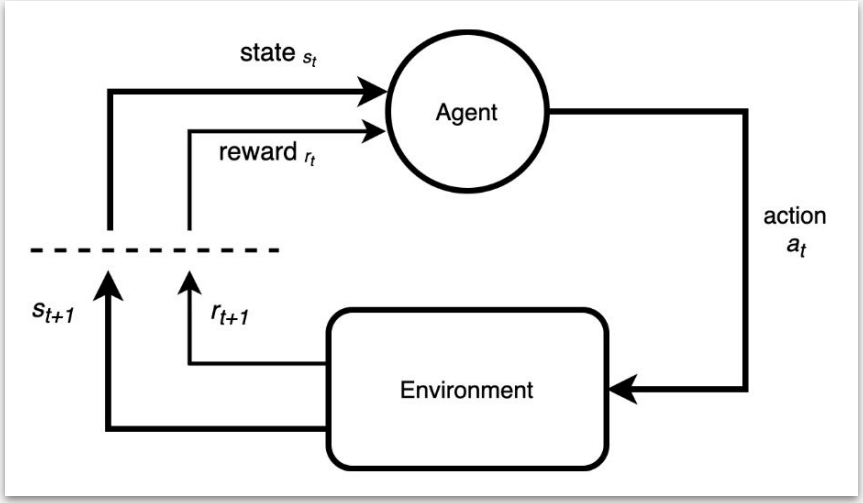
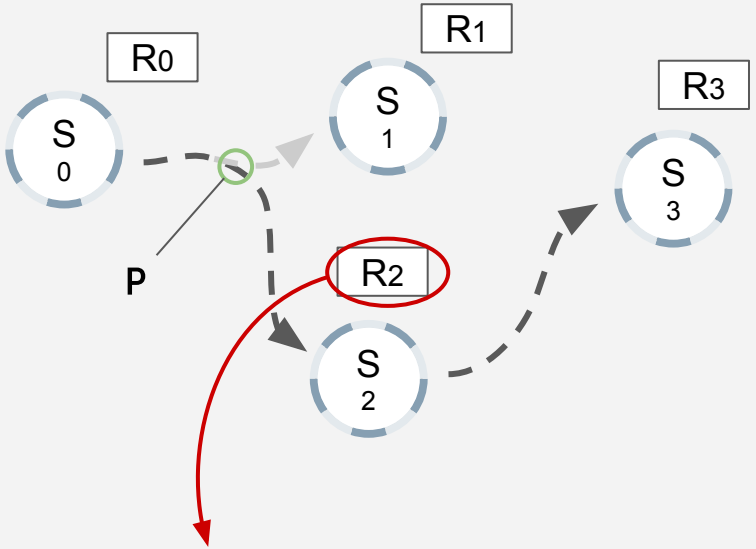
Reinforcement Learning and The Markov Decision Process

$$< S, A, P, R, \gamma >$$

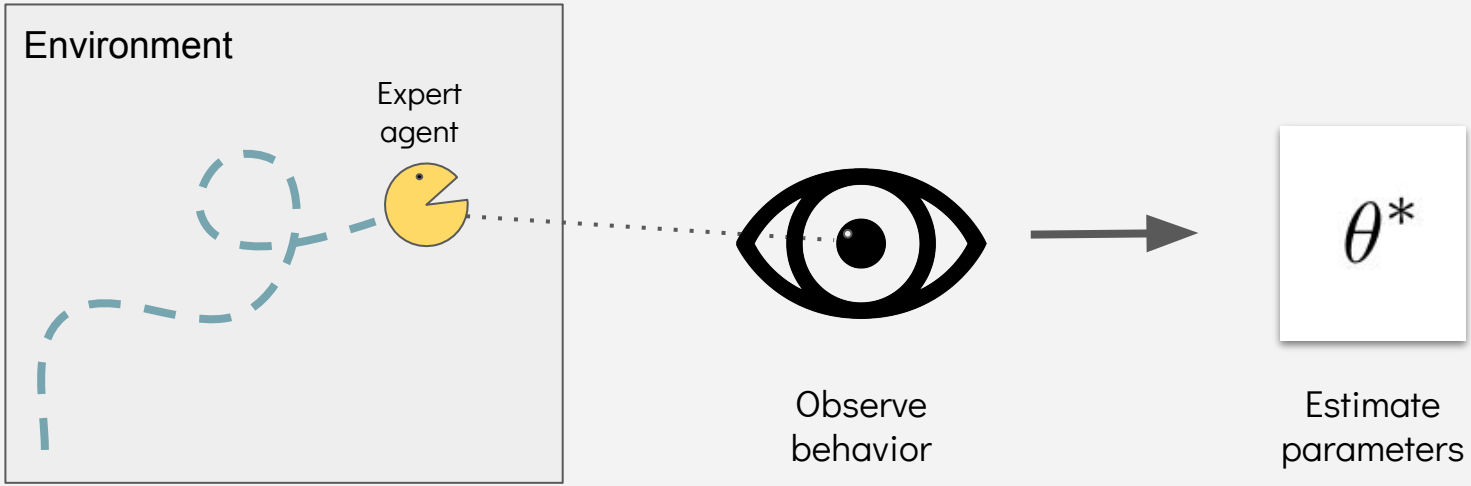




$$R(s,a) = \sum_i \omega_i * r_i$$

Inverse Reinforcement Learning

The Inverse problem of Reinforcement Learning consists in **inferring parameters** that characterize an agent or the environment by **observing its behaviour** in the given environment.



Inadequacy of the Expected Return

A **risk neutral agent** will behave as to maximise the expected value of return over a trajectory.

$$max\left(\mathbb{E}\left[\sum_t R_t\right]\right)$$

This **does not adapt well** to environments that include unrecoverable states.

Risk-neutrality does not reflect **human behaviour** .

Accounting for risk in RL - Inherent Uncertainty

Modified utility function

$$\text{ERM} = \frac{1}{\lambda} log \, \mathbb{E}_{\pi}[exp(-\lambda G)]$$

Risk Measures

$$\text{Return variance} = Var[\sum_{t} R_t]$$

$$\text{Reward volatility} = \mathbb{E}[\sum_{t} (R_t - J)^2]$$

GOAL

Our goal

Through the **observation** of a trained agent, acting optimally with respect to some risk-sensitive objective function, we want to:

- Estimate the **level of risk sensitivity** of the agent.
- Estimate the **weights** that characterize the environment's **reward** function.

APPROACH

Gradient-based Inverse Reinforcement Learning (GIRL)

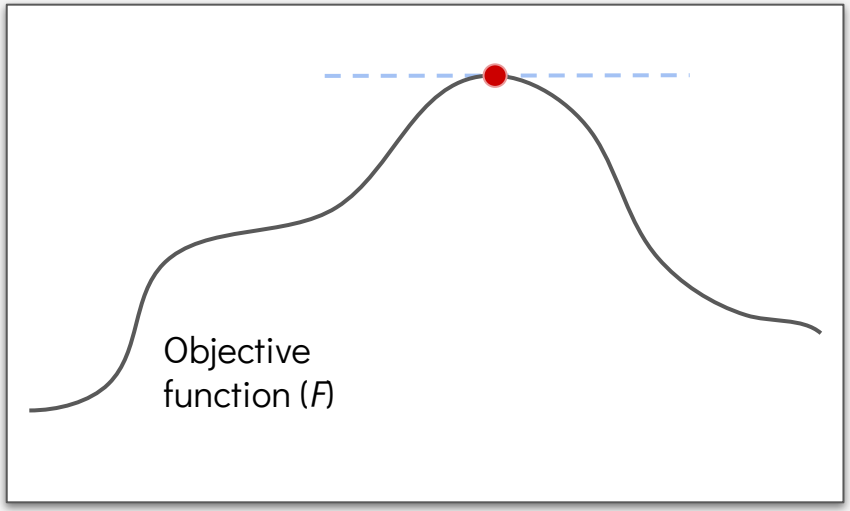
IRL trough Policy Gradient Minimization - [Pirotta; Restelli. 2016]

IRL through minimization of gradient norms

The expert's reward is a stationary point of its objective function.

$$\nabla_\theta F(\pi_\theta^E, R(\underline{\boldsymbol{\omega}^E})) = 0$$

$$\boldsymbol{\omega}^E = \underset{\boldsymbol{\omega}}{\operatorname{argmin}} \left\| \nabla_\theta F(\pi_\theta^E, R(\boldsymbol{\omega})) \right\|_2^2$$



Gradient-based Inverse Reinforcement Learning (GIRL)

[Pirotta, Restelli]

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$$\boldsymbol{\omega}^E = \underset{\boldsymbol{\omega}}{\operatorname{argmin}} \underbrace{\|\nabla_\theta F(\pi_\theta^E, R(\boldsymbol{\omega}))\|_2^2}_{g}$$

$$\boldsymbol{\omega}^{\boldsymbol{E}} = \underset{\boldsymbol{\omega}}{\operatorname{argmin}} \sum_j g(\theta_j, \boldsymbol{\omega})^2$$

$$\frac{\partial}{\partial \omega_l} f(\boldsymbol{\theta}, \omega_l) = \sum_j 2g(\theta_j, \boldsymbol{\omega}) \frac{\partial}{\partial \omega_l} g(\theta_j, \omega_l)$$

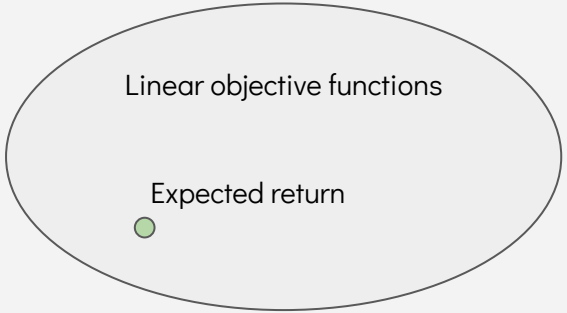
Risk-Averse GIRL

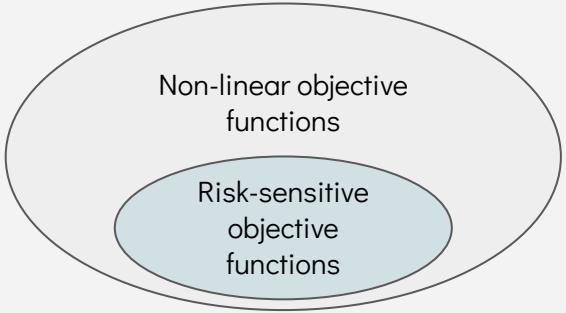
We extend the GIRL algorithm to the non-linear space.

$$F = \mathbb{E}[\sum_t R_t]$$

$$F = \underline{U}(R)$$

non-linear





Risk-Averse GIRL

The objective functions we implemented.

$$\text{ERM} = \frac{1}{\lambda} log \mathop{\mathbb{E}}_{\pi}[exp(-\lambda G)]$$

$$\text{Mean-Volatility} = \mathop{\mathbb{E}}_{\pi}[G] - \lambda \mathop{\mathbb{E}}_{\substack{s \sim d_{\mu,\pi} \\ s \sim \pi(\cdot|s)}} \left[(R(s,a) - J_\pi)^2\right]$$

$$\text{Mean-Variance} = \mathop{\mathbb{E}}_{\pi}[G] - \lambda Var[G]$$

RA-GIRL: Entropic Risk Measure

IRL through minimization of gradient norms

The expert's reward is a stationary point of its objective function.

$$G_{ERM} = -\frac{1}{\lambda} e^{-\lambda(R - \Psi_\lambda(\pi_\theta))}$$

$$\Psi_\lambda(\pi_\theta) = -\frac{1}{\lambda} log\, \mathbb{E}[exp(-\lambda G)]$$

$$\boldsymbol{\omega}^E = \underset{\boldsymbol{\omega}}{\operatorname{argmin}} \|\nabla_\theta J_{ERM}\|_2^2$$

RA-GIRL: Mean-volatility

IRL through minimization of gradient norms

The expert's reward is a stationary point of its objective function.

$$R_{VOLA} = R_{t'} - \lambda \frac{1-\gamma}{1-\gamma^T}(R_{t'} - J)^2$$

$$\boldsymbol{\omega}^E = \underset{\boldsymbol{\omega}}{\operatorname{argmin}} \|\nabla_\theta J_{VOLA}\|_2^2$$

RA-GIRL: Mean-variance

IRL through minimization of gradient norms

The expert's reward is a stationary point of its objective function.

$$\begin{aligned}\text{Mean-Variance} &= J - \lambda Var[G] \\ &= J - \lambda(M - J^2)\end{aligned}$$

$$\boldsymbol{\omega}^E = \underset{\boldsymbol{\omega}}{\operatorname{argmin}} \|\nabla_\theta J_\theta - \lambda(\underline{\nabla_\theta M - 2J\nabla_\theta J})\|_2^2$$

Variance Adjusted AC Algorithms - [Tamar et al. 2013]

Closed-form estimation of lambda

For the **mean-variance** and **mean-volatility** objective functions, we derived the closed-form equations for lambda, when the reward weights are known.

$$\min_{\lambda} ||\nabla_{\theta} L(\theta, \lambda)||_2^2 = \min_{\lambda} \sum_{d} (\nabla_{\theta_d} J(\theta) - \lambda \nabla_{\theta_d} V(\theta))^2$$

$$\implies \lambda = \frac{\sum_d \nabla_{\theta_d} J(\theta) \cdot \nabla_{\theta_d} V(\theta)}{||\nabla^2_{\theta_d} V(\theta)||}$$

Risk-Averse GIRL

Key advantages of RA-GIRL

- Does not require knowledge of the **model**.
- Can operate with a given **batch** of trajectories.
- Does not need to solve the **forward problem**.

Other approaches

Risk Sensitive IRL via Semi and Non-Semi Parametric Methods

(Singh et al. 2018)

Gradient-Based Inverse Risk-Sensitive Reinforcement Learning

(Mazumdar et al. 2017)

Risk-Averse GIRL

Coherent risk measures (CRM) expressed as **worst case** scenario for expected return over parameter uncertainty within a **risk envelope** .

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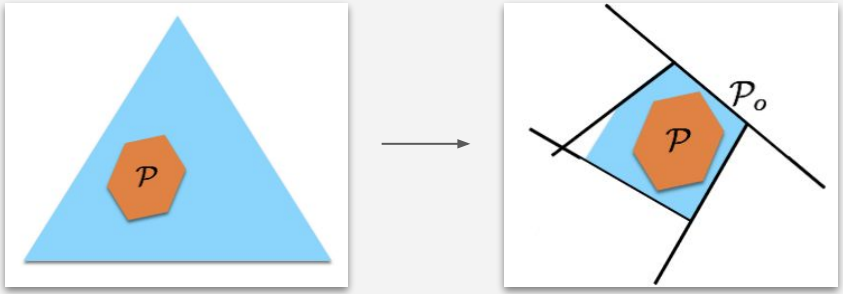

- ✗ Does not generalise well to step-wise uncertainty.
- ✗ Restricted to CRMs.

Risk-Averse GIRL

Other approaches

Risk Sensitive IRL via Semi and Non-Semi Parametric Methods

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Gradient-Based Inverse Risk-Sensitive Reinforcement Learning

(Mazumdar et al. 2017)

Gradient based IRL algorithm that minimizes a loss function over observed behaviour.

- Estimate policy that induces same distribution over state visits.
- Estimate parameters that define the level of risk aversion of the expert.

✗ Requires prior knowledge of the model.

✗ Does not estimate reward function.

EMPIRICAL EVALUATIONS

Environment 2: S&P 500 Trading

Trading system that runs on **historical data** collected from the S&P 500 index from 1980 to 2019.

Short-selling

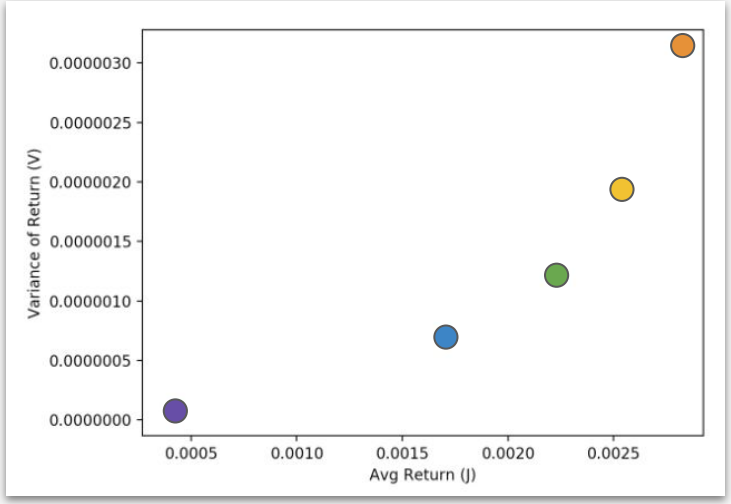
Going Flat

Long-selling

Environment 2: S&P 500 Trading: Mean-Variance

$$\text{Mean-Variance} = \mathop{\mathbb{E}}_{\pi}[G] - \lambda Var[G]$$

Mean-Variance Pareto frontier



Levels of risk-sensitivity

$\lambda = 1$

$\lambda = 4$

$\lambda = 6$

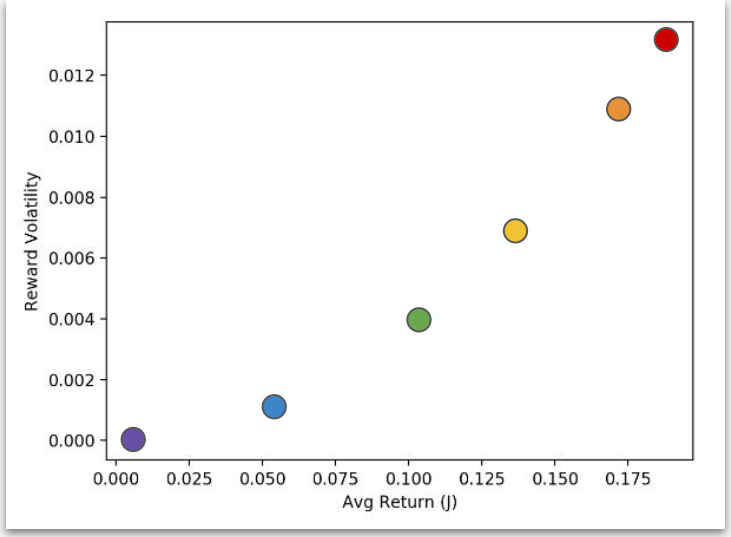
$\lambda = 20$

$\lambda = 100$

Environment 2: S&P 500 Trading: Mean-Volatility

$$\text{Mean-Volatility} = \mathop{\mathbb{E}}_{\pi}[G] - \lambda \mathop{\mathbb{E}}_{\substack{s \sim d_{\mu,\pi} \\ s \sim \pi(\cdot|s)}} \left[(R(s,a) - J_\pi)^2 \right]$$

Mean-volatility Pareto frontier



Levels of risk-sensitivity

$\lambda = 10$

$\lambda = 20$

$\lambda = 27$

$\lambda = 30$

$\lambda = 50$

$\lambda = 130$

Environment 2: S&P 500 Trading

$$R_t = P_t \frac{price_t - price_{t-1}}{price_{t-1}} - f|P_t - P_{t-1}|$$

$$R_{norm_t}(\boldsymbol{\omega}) = \omega_1(n * P_t \frac{price}{price_{t-1}}) - \omega_2(n * P_t) - \omega_3|P_t - P_{t-1}|$$

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Environment 2: S&P 500 Trading

Trading Mean-Variance

Weights Configuration	Expected Values	Estimated Values	Gradients L2-norm
Weight 1	0.499982501	0.5004	0.02465
Weight 2	0.499982501	0.4996	
Weight 3	3.50e-05	4.23-18	
λ	1	0.0	
	0.499982501	0.5002	0.04384
	0.499982501	0.4997	
	3.50E-05	13.65e-5	
	4	7.5232	
	0.499982501	0.5002	0.03020
	0.499982501	0.4998	
	3.50E-05	8.47e-22	
	6	15.2795	
	0.499982501	0.5001	0.01984
	0.499982501	0.4999	
	3.50E-05	0.0	
	20	21.2222	
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	3.50E-05	2.08e-4	
	100	2.5507	

Trading Mean-Volatility

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Weight 1	0.499982501	0.4994	2.4e-7
Weight 2	0.499982501	0.4973	
Weight 3	3.50e-05	3.26e-3	
λ	10	0.0837	
	0.499982501	0.5009	3.5e-10
	0.499982501	0.4991	
	3.50E-05	4.07e-23	
	20	5.8117	
	0.499982501	0.5002	1.25e-9
	0.499982501	0.4993	
	3.50E-05	4.72e-4	
	27	6.8618	
	0.499982501	0.5036	3.5e-17
	0.499982501	0.4963	
	3.50E-05	3.31e-5	
	30	17.9961	
	0.499982501	0.5017	3.0e-16
	0.499982501	0.4976	
	3.50E-05	7.18e-4	
	50	29.5076	
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Conclusions

Explored a novel approach to simultaneously estimate the **level of risk sensitivity** of an expert and the **weights of the reward function** being optimized that:

- Does not require knowledge of the **model**.
- Can operate with a given **batch** of trajectories.
- Does not need to solve the **forward problem**.

In addition, we provide closed form equations to accurately estimate the level of risk aversion when the reward function is known.

THANK YOU.

Slides bibliography:

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